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Channel Confinement: A Novel Approach to Tackle Batch Variation in Conjugated Polymers for Organic Electrochemical Transistors

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Polymer electronics have attracted extensive attention due to their intrinsic flexibility, structural modifiability, and cost-effective fabrication. However, compared to silicon electronics, polymer semiconductors suffer from their inherent polydispersity, resulting in variations between batches, which becomes a crucial challenge in polymer-based electronics and hinders their large-scale applications. In this study, the focus is on polymers that are significantly affected by batch variations and molecular weight effects in organic electrochemical transistors (OECTs). By introducing channel confinement effect, specifically by reducing the channel length (*L*) in vertical OECTs, molecular bridges are formed, leading to excellent immunity to molecular weight variation and remarkably enhanced device performance. OECTs based on n-type poly(benzimidazobenzophenanthroline) (BBL) and p-type poly[2,6-(4,4-bis-potassium butanylsulfonate-4H-cyclopenta-[2,1-b;3,4-b']-dithiophene)-alt-4,7-(2,1,3-benzothiadiazole)] (CPE-K) with varying molecular weights exhibit nearly identical and high transconductance of 75.00 mS for BBL and 120.00 mS for CPE-K, respectively, with an *L* of 35 nm. This phenomenon is elucidated by high-resolution transmission electron microscopy and geometry optimization. Furthermore, organic complementary inverters exhibit consistent and stable voltage gain of 60.00 V V^{-1} across BBL with different molecular weights and have been successfully used to amplify multiple types of physiological electrical signals. This strategy is expected to accelerate the large-scale production of polymer electronics.

1. Introduction

Inorganic semiconductors, such as monocrystalline silicon, rely on their ordered and repetitive lattice structures to ensure the reproducibility, accelerating their commercialization (**Figure 1a**).[1] In comparison with inorganic semiconductors, polymer semiconductors (PSCs) exhibit distinct advantages such as structural designability, solvent processability, and mechanical flexibility.[2–5] However, the inherent polydispersity of polymers leads to batch-to-batch variations, which significantly inhibit their large-scale industrialization (Figure 1b).[6,7] Organic mixed ionic-electronic conductors (OMIECs), as channel materials of organic electrochemical transistors (OECTs), represent a unique class of organic semiconductors that utilize strong ionic-electronic coupling to facilitate charge transfer and signal conversion, and have garnered substantial attention in the field of organic electronics.[8–11] Among them, some representative polymer-based OMIECs are also significantly influenced by batch-to-batch variations and molecular weight effect.[12–16]

To address this inherent problem of polymers, a series of solutions has been proposed (Figure 1c). From the synthetic pathway, by incorporating a low-temperature stage that facilitates the preferential growth of the low-molecular-weight segment, staged-heating Stille coupling has been reported for synthesizing ultrahigh-quality conjugated polymers with high molecular weight (M_w) and narrow $Đ$.[17] Homogeneous Suzuki-Miyaura coupling undergoes selective direct transmetalation through a boronate-base intermediate at room temperature, avoiding self-coupling defects and obtaining high-quality PSCs.[18] The iterative

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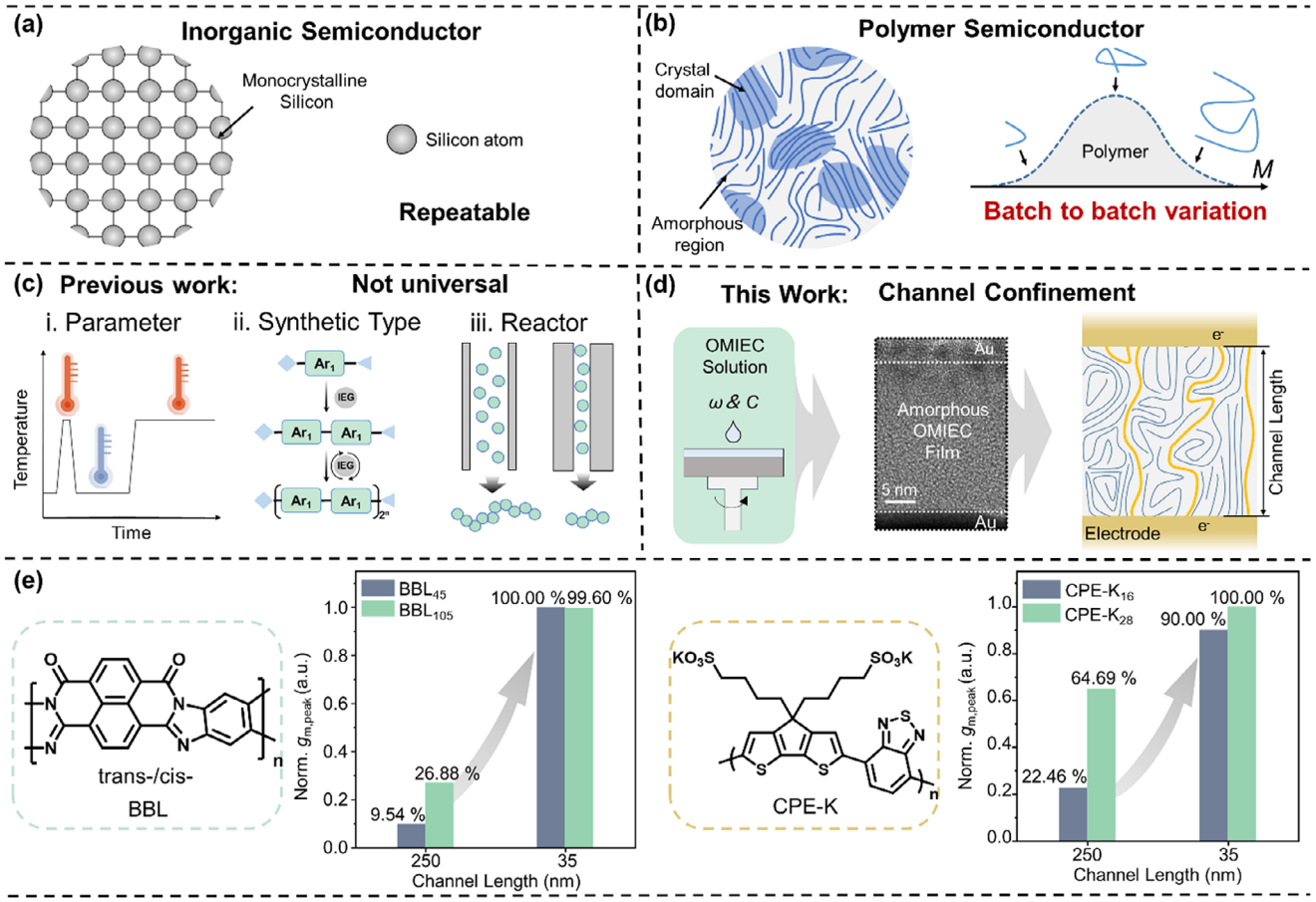


Figure 1. Schematic diagram of a) a silicon-based inorganic semiconductor and b) a polymer semiconductor. c) Description of previous studies on the control of batch-to-batch variation. d) Universal channel confinement strategy to tackle batch-to-batch variation described in OECTs. e) The trend in batch-to-batch variation exhibited by BBL and CPE-K with different molecular weights in OECTs with varying channel lengths was investigated. The performance of these devices has been normalized based on the maximum transconductance.

exponential growth strategy offers an effective method for synthesizing macromolecules while enabling precise control over chain length, sequence, and stereoconfiguration, making it a promising approach for achieving precise control of conjugated polymers.[19] Furthermore, nanoscale reactors have been employed to achieve precise regulation of polymer structures.[20] The above methods either rely on cross-coupling reactions with highly selective reactive groups or require unconventional reaction equipment, resulting in limited applicability. Moreover, for poly(benzimidazobenzophenanthroline) (BBL)[21] and poly[2,6-(4,4-bis-potassium butanylsulfonate-4H-cyclopenta-[2,1-b;3,4-b']-dithiophene)-alt-4,7-(2,1,3-benzothiadiazole) (CPE-K),[22] whose performance in OECTs is significantly influenced by molecular weight effect, these aforementioned methodologies are not applicable. Therefore, it is essential to explore a novel and universal approach that can enhance the batch-to-batch repeatability in the subsequent large-scale manufacturing of OECTs.

The performance of an OECT is primarily determined by the figure-of-merit (μC^*), where μ represents the charge carrier mobility and C^* denotes the volumetric capacitance of the active material.[23] Previous studies have demonstrated the influence of channel length on charge transport and ionic penetration in OMIECs but lacked systematic investigations across different molecular weights.[24,25] In our study, it is found that for a given type of polymer, the influence of molecular weight on the C^* can be considered negligible. Therefore, the molecular weight effect is attributed primarily to differences in charge transport rather than ionic storage.

Charge transport theory of organic electronics is fundamental to understanding both material conductivity and device performance. The electron transport mechanisms in organic semiconductors are typically categorized into two principal models. Band-like transport is used to describe strong crystalline packing order materials, while it is more reasonable to use hopping transport to describe low-crystalline and amorphous ones.[26,27] Charge carriers primarily move via intramolecular conduction and intermolecular π–π stacking. However, when molecular packing is insufficiently ordered, carriers must "hop" between localized states through thermal excitation, resulting in limited mobility.[28] This issue is more pronounced in low-molecular-weight polymers. Therefore, we assume that the probability of charge hopping exhibits a correlation with both molecular chain length and channel length (L). Based on this, the channel confinement effect was proposed.

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Here, by employing a channel confinement strategy, highly consistent OECT performances are obtained in both n- and p-type OMIECs with different M_w, significantly mitigating batch-to-batch variations and molecular weight effect.[14–16] Specifically, when L is reduced to 35 nm by adopting vertical OECTs (vOECTs) architecture (Figure 1d),[25] OECTs based on BBL and CPE-K with different M_w exhibit nearly identical peak transconductance ($g_{m,peak}$) of 75.00 mS (drain voltage (V_D) = 0.50 V, gate voltage (V_G) = 0.70 V) and 115 mS (V_D = −0.50 V, V_G = −0.70 V), and on current (I_{ON}) of 30.00 mA and 45.00 mA, respectively (Figure 1e). While a transistor with longer L shows tremendous performance discrepancy due to the M_w variation. Therefore, the channel confinement effect is proposed. When L is much greater than the molecular chain length in low-crystalline PSCs, carriers are transported between molecules via hopping mechanisms. Conversely, when L is reduced to the scale of the molecular chain length, the polymer chains are able to directly bridge the electrodes, enabling carrier transport via intrachain conduction pathways. This effect is supported by high-resolution transmission electron microscopy (HRTEM) and geometric optimization calculations. Furthermore, vertically stacked complementary inverters based on BBL and gDPP-g2T[25] based vOECTs are fabricated and demonstrate negligible batch-to-batch variation with a stable voltage gain of ≈60.00 V V^{-1} under driving voltage of 0.60 V. These findings highlight the effectiveness of channel confinement in overcoming molecular weight effect and batch-to-batch variations, offering a robust strategy for achieving high-performance OECTs with low-molecular-weight polymers.

2. Results and Discussion

2.1. Materials Synthesis

The synthesis of BBL and CPE-K followed a previously reported method.[14,29] Detailed synthetic procedures are described in the Schemes S1 and S2 (Supporting Information). The viscosity-average molecular weight (M_v) of five BBL batches is measured as 15.4, 20.5, 25.5, 29.5, and 34.5 kDa, corresponding to degrees of polymerization of 45, 60, 75, 90, and 105, respectively, they are referred to as BBL_{45}, BBL_{60}, BBL_{75}, BBL_{90}, and BBL_{105} (Figure S1, Supporting Information). The weight-average molecular weight (M_w) of three CPE-K batches is measured as 10.3, 13.4, and 18.3 kDa, corresponding to degrees of polymerization of 16, 21, and 28, respectively, and they are referred to as CPE-K_{16}, CPE-K_{21}, and CPE-K_{28} (Figure S2, Supporting Information). Three Pg2T-T variants with distinct molecular weights are synthesized and designated as Pg2T-T_L, Pg2T-T_M, and Pg2T-T_H based on GPC results (Figure S3, Supporting Information). By controlling concentration and spin coating speed, the thicknesses of BBL and CPE-K films ranging from 35 to 250 nm are precisely fabricated (Figures S4 and S5, Supporting Information).

2.2. OECT Performances

OECT performances with varying M_w are characterized to investigate the impact of L. As a reference, pOECTs (L = 10 µm) based on all BBL and CPE-K are fabricated first (Figure S6, Supporting Information), along with the characterization of corresponding transfer and output characteristics (Figures S7 and S8, Supporting Information). For BBLs, as the M_v enhances, the average I_{ON} (V_D = 0.50 V, V_G = 0.70 V) of pOECTs monotonously increases from 11.20 µA (BBL_{45}) to 30.90 µA (BBL_{105}). Similarly, the $g_{m,peak}$ gradually increases from 48.80 µS (BBL_{45}) to 114.00 µS (BBL_{105}), along with a natively shifted turn-on voltage (V_{ON}) from 0.010 V (BBL_{45}) to -0.016 V (BBL_{105}). For CPE-Ks, as the M_w enhances, the average I_{ON} (V_D = −0.50 V, V_G = −0.70 V) monotonously increases from 10.90 µA (CPE-K_{16}) to 172.00 µA (CPE-K_{28}). Similarly, the $g_{m,peak}$ gradually increases from 29.10 µS (CPE-K_{16}) to 464.00 µS (CPE-K_{28}), along with a natively shifted V_{ON} from −0.159 V (CPE-K_{16}) to −0.023 V (CPE-K_{28}). Such a trend is consistent with the previous report, where OECTs show enhanced performance with higher M_v.[16,30]

On the other hand, according to Bernard's formula, the performance of the transistor can be improved with a shorter L.[31–33] Since it is challenging to achieve L below 1 µm in a planar structure using conventional photolithography equipment,[34,35] electron beam lithography (EBL)[36–38] is utilized for BBL_{75}-based pOECTs with submicron L (Figures S9 and S10, Supporting Information). Specifically, as L decreases from the micron scale to the nanometer scale, I_{ON} increases from 1.67 × 10^{-2} mA to 7.74 × 10^{-2} mA, and the $g_{m,peak}$ gradually increases from 6.00 × 10^{-2} to 2.10 × 10^{-1} mS as L decreases from 10 µm to 200 nm (V_D = 0.50 V, V_G = 0.70 V), respectively. However, all short-channel pOECTs show unstable characteristics after repeated testing and become impaired after about five transfer characteristic tests (Figure S11, Supporting Information). It is speculated that these unusual characteristics, along with the rapid decay, are attributed to strong film swelling with short channels and the un-patterned nature of the BBL films. On the other hand, the high-precision EBL fabrication process is time-consuming, complex, and shows low yield, which greatly reduces research efficiency.

In vOECTs, OMIECs are sandwiched between two layers of Au electrodes through a two-step evaporation process, with the L defined by the film thickness (**Figure 2a**). Consequently, five different M_v BBL-based vOECTs are fabricated with L of 250, 200, 150, 100, and 35 nm, respectively, while maintaining a consistent channel area of 45 µm × 45 µm. Detailed experimental procedures are provided in the supporting information (Figure S12, Supporting Information). Transfer and output characteristic curves of all those devices are characterized thereafter (Figure 2b,c; Figures S13–S18, Supporting Information). As L decreases, the hysteresis between forward and backward transfer curves of vOECTs at all M_w reduces significantly, indicating enhanced reversibility and faster ion motion in the electrochemical doping/dedoping process. When employing BBL_{45} as the channel, as L decreases from 250 to 35 nm, I_{ON} and $g_{m,peak}$ increases from 2.76 mA and 7.24 mS to 28.60 mA and 75.90 mS, respectively (V_D = 0.50 V, V_G = 0.70 V). In addition, as L decreases from 250 to 35 nm, the V_{ON} also shows a significant decrease from −0.205 to and −0.245 V, indicating that the process of ionic doping is relatively facile to achieve. Similar trends are also observed in vOECTs based on BBL with other M_v, where $g_{m,peak}$, I_{ON}, and V_{ON} exhibit improved performance as L decreases (Figure 2d; Figures S19 and S20, Supporting Information). All OECT performances are extracted and summarized in Tables S1 and S2 (Supporting Information).

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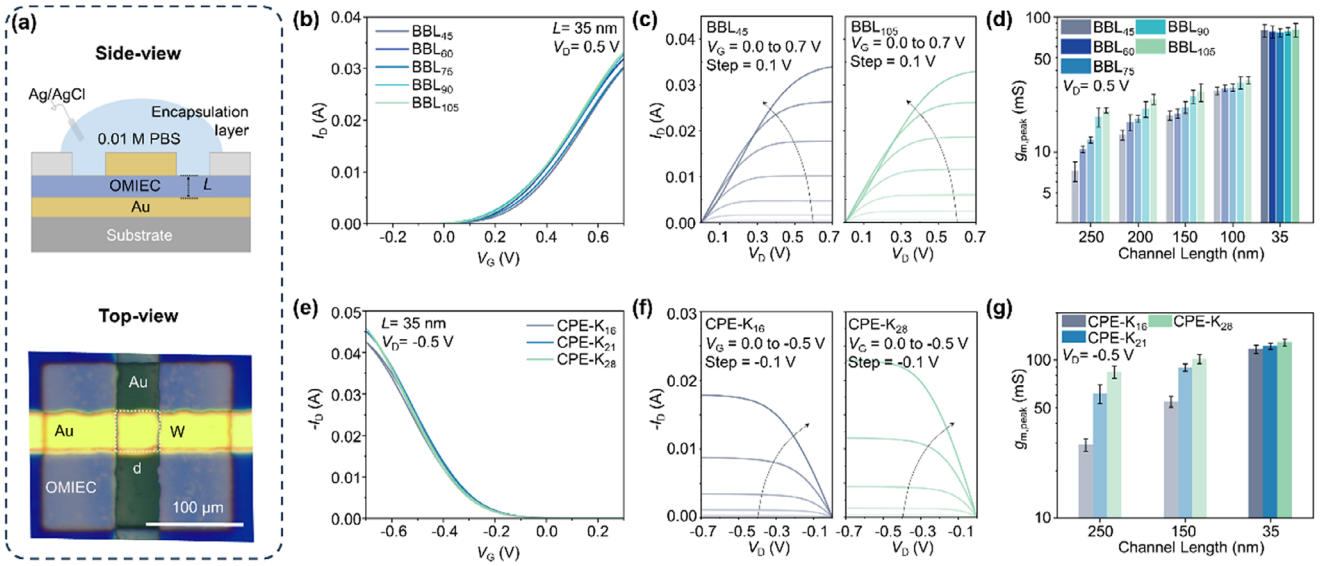


Figure 2. OECT performances based on BBL and CPE-K with different M_v. a) Side-view schematics and top-view optical photograph (L = 35 nm) of vOECTs. b,e) Comparison of the transfer curves of b) BBL and e) CPE-K with different molecular weights under a 35 nm channel length. c,f) Output curves of c) BBL_{45} and BBL_{105} based vOECTs and f) $CPE\text{-}K_{16}$ and $CPE\text{-}K_{28}$ based vOECTs with L of 35 nm. d,g) Peak transconductance ($g_{m,peak}$) of vOECTs with L ranging from 250 to 35 nm based on d) BBL and g) CPE-K with different M_v. All experimental data are collected from at least eight samples.

It is observed that as the L decreases, the gap in $g_{m,peak}$ among different M_v gradually narrows until it becomes negligible. Specifically, when L = 250 nm, I_{ON} and $g_{m,\,peak}$ of BBL_{45} and BBL_{105} (V_D = 0.50 V, V_G = 0.70 V) exhibit obvious differences of 4.77 mA and 13.16 mS, respectively (percentage changes are 172.80% and 181.70%, respectively). In contrast, as L is reduced to 35 nm, the differences of I_{ON} and $g_{m,\,peak}$ reduce to 3.10 mA and 0.30 mS, respectively (percentage changes are 10.80% and 0.40%, respectively). BBL-based vOECTs with L of 35 nm demonstrated the highest performance in all the fabricated OECTs, achieving I_{ON} of ≈30.00 mA and $g_{m,peak}$ of ≈74.00 mS. In comparison to pOECTs, vOECTs with L of 35 nm exhibit significantly higher I_{ON}, $g_{m,peak}$, and I_{ON}/I_{OFF}, which are about three orders of magnitude higher than those on pOECTs counterparts (Figure S21, Supporting Information; **Table 1**). Moreover, enhanced cycling stability (10 000 cycles, over 80% current remained) is also achieved (Figure S22, Supporting Information).

To verify batch repeatability in short-channel OECTs, BBL-based vOECTs with a L of 35 nm are fabricated on both rigid (Si/SiO_2) and flexible (polyimide, PI) substrates. The transfer and output curves of vOECTs are analyzed for each BBL batch. Specifically, 20 devices are uniformly selected for testing on two 2 cm × 2 cm large-area vOECTs, yielding a total of 40 data points to evaluate performance deviations within each batch. Excellent reproducibility is demonstrated across random geometric positions of the whole devices, with relatively small standard deviations (Figures S23 and S24, Supporting Information). For BBL_{45}, BBL_{105}, $BBL_{45/105,}$ and BBL_{75}, the I_{ON} (V_D = 0.50 V, V_G = 0.70 V) is 27.84, 28.92, 27.61, and 27.33 mA, while the $g_{m,peak}$ is 73.32, 74.68, 72.94, and 74.22 mS, respectively. Additionally, there is no significant difference in performance between Si/SiO_2 and PI substrates. For example, when BBL_{45} is used as the channel, the I_{ON} is 26.39 mA and the $g_{m,peak}$ is 76.07 mS, respectively. Similar trends are observed across

Table 1. Performance summary of vOECTs and pOECTs prepared by BBL and CPE-K with different molecular weights.

Materials[a]	vOECT L = 35 nm				pOECT L = 10 μm			
	V_{ON} [V]	I_{ON} [mA]	$g_{m,peak}$ [mS]	$I_{ON/IOFF}$	V_{ON} [V]	I_{ON} [μA]	$g_{m,peak}$ [μS]	$I_{ON/IOFF}$
BBL_{45}	−0.218 ± 0.023	28.6 ± 1.6	75.9 ± 2.6	1.24 ± 9.23 × 10^6	0.010 ± 0.019	11.2 ± 4.8	48.8 ± 6.6	1.19 ± 0.62 × 10^2
BBL_{60}	−0.236 ± 0.012	30.2 ± 1.4	73.6 ± 2.5	1.02 ± 5.28 × 10^6	0.010 ± 0.011	13.8 ± 2.2	52.3 ± 7.3	1.29 ± 0.33 × 10^2
BBL_{75}	−0.236 ± 0.005	30.0 ± 1.7	76.1 ± 2.5	6.43 ± 2.18 × 10^6	−0.014 ± 0.016	16.7 ± 3.2	60.0 ± 6.6	1.80 ± 0.75 × 10^2
BBL_{90}	−0.234 ± 0.007	31.9 ± 1.1	75.8 ± 2.1	3.29 ± 1.83 × 10^6	−0.020 ± 0.010	22.7 ± 2.4	81.1 ± 5.8	8.46 ± 5.08 × 10^1
BBL_{105}	−0.239 ± 0.007	31.7 ± 1.9	75.6 ± 2.6	5.85 ± 2.19 × 10^5	−0.016 ± 0.014	30.9 ± 5.9	114.0 ± 17.0	2.21 ± 1.10 × 10^1
$CPE\text{-}K_{16}$	0.244 ± 0.012	42.4 ± 3.7	117.0 ± 7.0	4.53 ± 0.99 × 10^4	−0.159 ± 0.014	10.9 ± 1.8	29.1 ± 3.8	2.01 ± 1.86 × 10^1
$CPE\text{-}K_{21}$	0.226 ± 0.015	44.8 ± 2.8	123.0 ± 5.0	2.91 ± 2.14 × 10^4	−0.053 ± 0.032	131.0 ± 19.8	360.0 ± 48.2	2.51 ± 1.72 × 10^1
$CPE\text{-}K_{28}$	0.235 ± 0.011	45.3 ± 3.2	130.0 ± 6.0	3.17 ± 0.49 × 10^4	−0.023 ± 0.031	172.0 ± 14.1	464.0 ± 46.0	2.12 ± 0.63 × 10^1

[a] All the results are based on eight valid statistics. For BBL, V_D = +0.5 V, and for CPE-K, V_D = −0.5 V.

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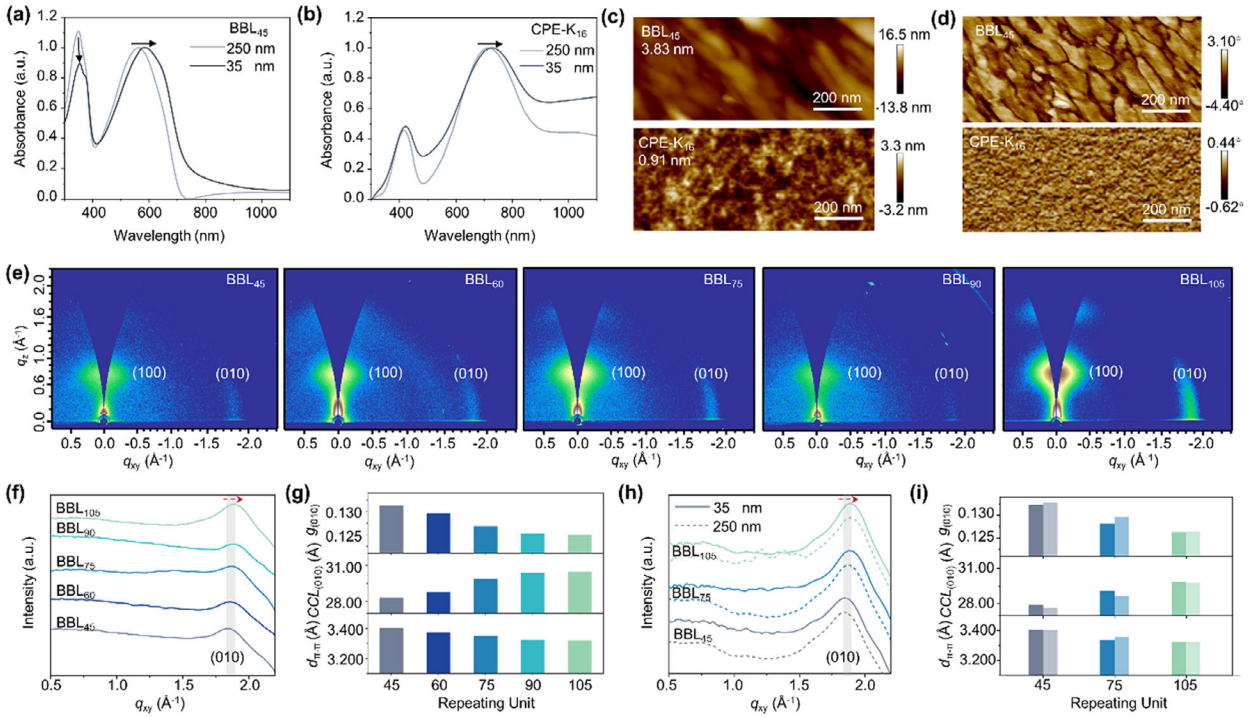


Figure 3. Physicochemical properties and film morphology characterization. a,b) UV–vis–NIR absorption spectra of 35 and 250 nm thin a) BBL_{45} and b) CPE-K_{16} films. c) AFM height images of BBL_{45} and CPE-K_{16} thin film. d) AFM phase images of BBL_{45} and CPE-K_{16} thin films. e) 2D GIWAXS patterns of BBL_{45}, BBL_{60}, BBL_{75}, BBL_{90} and BBL_{105} 35 nm thin films. f) In-plane line cuts show the shift of (010) scattering peaks in 35 nm thin films. g) Variation trends in $d_{\pi-\pi}$, $CCL_{(010)}$ and $g_{(010)}$ for BBL_{45}, BBL_{60}, BBL_{75}, BBL_{90} and BBL_{105} 35 nm thin films. h) In-plane line cuts show the shift of (010) scattering peaks in 35 and 250 nm thin films. i) Variation trends in $d_{\pi-\pi}$, $CCL_{(010)}$ and $g_{(010)}$ for BBL_{45}, BBL_{75}, and BBL_{105} 35 and 250 nm thin films.

other BBL batches, and all results are presented in the supporting information. These findings demonstrate the strategy's robust reproducibility and its potential applicability to various substrates.

Subsequently, CPE-K-based vOECTs are fabricated at L of 35, 150, and 250 nm. Transfer and output characteristic curves of all those devices are characterized thereafter (Figure 2e–g; Figures S25–S28, Supporting Information). When employing CPE-K_{16} as the channel, as L decreases from 250 to 35 nm, I_{ON} and $g_{m,peak}$ (V_D = −0.50 V, V_G = −0.70 V) increase from 9.41 mA and 29.20 mS to 31.00 mA and 84.10 mS, respectively, while the gap in $g_{m,peak}$ among different M_v gradually narrows. Specifically, when L = 250 nm, I_{ON} and $g_{m,peak}$ of CPE-K_{16} and CPE-K_{28} exhibit obvious differences of 21.59 mA and 54.90 mS, respectively (percentage changes are 229.44% and 188.00%, respectively). In contrast, as L is reduced to 35 nm, the differences of I_{ON} and $g_{m,peak}$ reduce to 2.90 mA and 13.00 mS, respectively (percentage changes are 6.84% and 11.11%, respectively). All OECT performances are extracted and summarized in Tables S3–S5 (Supporting Information).

To conclude, despite increasing M_v in vOECTs (L = 35 nm), I_{ON} and $g_{m,peak}$ remain approximately constant rather than increasing with M_v. Among all M_w-related OECTs reports, this is the first one in which the device performance is not affected by it (Figure S33, Supporting Information). OECTs with low M_v can still achieve the same high performance as those with a high molecular weight, with L reducing to 35 nm. Furthermore, compared to pOECTs, vOECTs can achieve higher and more stable performance, along with negligible hysteresis and enhanced stability.

2.3. Physicochemical Properties and Film Morphology Characterization

To investigate the reason behind the significant disappearance of batch-to-batch variations when the channel length (L) is reduced to 35 nm, the physicochemical properties and microstructure of BBL and CPE-K films are characterized. UV–vis-NIR spectroscopy measurements are carried out to probe the optical properties of both solution and thin-film states of BBL and CPE-K (**Figure 3**a,b; Figure S34, Supporting Information).

For BBLs, all curves exhibit two absorption peaks: peak I for n-π^* transition absorption ≈300–400 nm and peak II for π–π^* transition absorption at 400–800 nm. At the same thickness, different batches of BBL exhibit a slight redshift ($\Delta\lambda \approx 5$ nm in 250 nm and 35 nm films). However, within a single batch such as BBL_{45}, a significant redshift ($\Delta\lambda \approx 20$ nm) in the absorption peak is observed as the film thickness decreases, accompanied by a marked reduction in the intensity ratio of peak I, suggesting improved molecular packing and aggregation. Additionally, the starting wavelengths across batches are similar (≈716 nm), indicating comparable optical bandgaps (E_{opt} = 1.74 eV). CPE-K samples display three absorption peaks: peak I for π–π^* transition absorption ≈300–500 nm, peak II for intramolecular charge transfer absorption at 500–900 nm, and peak III for polaron

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absorption at 1150 nm. At a fixed thickness, high-molecular-weight CPE-K exhibits a significant redshift ($\Delta\lambda$ ≈30 nm in 250 nm films and 20 nm in 35 nm films) along with a noticeably stronger polaron absorption. Within a single batch, such as CPE-K_{16}, film thinning results in an additional redshift ($\Delta\lambda$ ≈16 nm). These results indicate that reducing the film thickness slightly enhances molecular aggregation. To explore the influence of film thickness on polarization ability, we measure the spectroelectrochemistry of BBL_{75} with different thicknesses (Figure S35, Supporting Information). The results of film thick-normalized differential spectroelectrochemistry show that the polarization ability is less affected by film thickness. As measured by cyclic voltammetry (CV), the lowest unoccupied molecular orbital energy level is estimated to be ≈−4.38 eV, and the maximum occupied molecular orbital energy level is estimated to be ≈−4.24 eV, while independent of molecular weight (Figure 3c; Figure S36, Supporting Information). Furthermore, AFM analysis reveals that spin-coated BBL and CPE-K films exhibit continuous and uniform morphology, with slight variations in root mean square roughness (σ_{RMS}) across different batches (e.g., 3.83 nm for BBL_{45}, 0.91 nm for CPE-K_{16}, respectively) (Figure 3d; Figures S37 and S38, Supporting Information). The influence of surface roughness on device performance is systematically investigated. vOECTs with 35 nm channel length, which are fabricated from BBL films with varying roughness, exhibited nearly identical electrical characteristics. These findings indicate that surface roughness is not a decisive factor in determining device performance (Figures S39–S42, Supporting Information).

Based on these tests, the physicochemical properties of BBL and CPE-K across different molecular weights and film thicknesses were consistent with trends reported.[12,16,39] To further investigate the microstructural characteristics of films, BBL was selected for grazing-incidence wide-angle X-ray scattering (GIWAXS) analysis, and all batches showed a distinct edge orientation. (Figure 3e–i) At a thickness of 35 nm, BBL films with M_v ranging from 15 to 35 kDa exhibit enhanced lamellar (100) diffraction peaks at q_z = 0.76–0.79 $Å^{-1}$ (corresponding d-spacing of 8.26–7.96 Å) and π–π stacking (010) peaks at q_{xy} of 1.85–1.89 $Å^{-1}$ (d-spacing of 3.40–3.32 Å), respectively. Three batches of BBL (BBL_{45}, BBL_{75}, and BBL_{105}) are selected for microstructure with film thicknesses of 35 nm and 250 nm. The results show that the film thickness has no significant effect on the molecular accumulation within the allowable error range, and the BBL with higher M_v still has stronger crystallinity and more ordered accumulation. (Figure S43 and Table S6, Supporting Information). In brief, higher M_v shows stronger π–π stacking, longer crystal coherence lengths, and lower paracrystalline disorder, all of which contribute to enhanced intermolecular charge mobility. However, the existing characterization of physicochemical properties and microscopic morphology of the thin film cannot explain the experimental phenomenon. It is necessary to employ an alternative methodology to better understand the impact of reduced L on OECT performance.

2.4. Establishment of Channel Confinement Effect

The electrochemical properties of BBL and CPE-K thin films of 35 and 250 nm are determined by electrochemical impedance test (EIS, **Figure 4a**; Figures S44 and S45, Supporting Information). Similar volumetric capacitances were observed across different batches of BBL (300–600 F cm^{-3}) and CPE-K (65–75 F cm^{-3}). In addition, the effect of contact resistance on OECT performance is considered (Figure 4b; Figure S46, Supporting Information). As L decreased from 250 to 35 nm, the R_{line} of all batches of BBL showed a significant decline. Specifically, the R_{line} of BBL_{45} decreased from 184.00 to 16.90 Ω, and that of BBL_{105} decreased from 66.60 to 15.20 Ω. Additionally, when L = 0 nm, the R_{line} is measured as 5.50 Ω, accounting for only about one-third of the total resistance at 35 nm. This indicates that the R_C is not the primary factor influencing I_{ON}. So, the enhancement in OECT performance resulting from the reduction in L is primarily attributed to the increase in charge carrier mobility.

The top view and cross-sectional HRTEM images of BBL and CPE-K thin are obtained. The results reveal that BBL films, regardless of molecular weight, exhibit weakly crystalline and uniformly distributed morphologies. Similarly, CPE-K also obtained a similar structure (Figure 4c,d; Figures S47 and S48, Supporting Information). It is worth noting that, during focused ion beam (FIB) preparation, separation between the gold top layer and the OMIEC film was observed, due to stress introduced during processing. In addition, gold particles were found near the top of the films, which resulted from the penetration of gold during the evaporation process. The elemental mapping of BBL and CPE-K is presented in Figures S49 and S50 (Supporting Information). Together, both characterization techniques consistently demonstrate that the two OMIEC materials possess weak crystallinity and are predominantly amorphous in the bulk phase. As a result, electron transport primarily occurs via interchain hopping, which inherently leads to pronounced molecular weight dependence and batch-to-batch variability over longer transport distances. Geometric optimization indicates that the molecular lengths of BBL_{45}, BBL_{60}, BBL_{75}, BBL_{90}, and BBL_{105} are 53.67, 71.49, 89.31, 107.13, and 124.95 nm, respectively. For CPE-K_{16}, CPE-K_{21}, and CPE-K_{28}, the molecular lengths are calculated as 19.58, 25.65, and 34.15 nm, respectively (Figure S51, Supporting Information). The correlation between the charge transport capability of polymers and M_w can be quantitatively described. However, these studies have ignored the influence of L.[40]

The confinement effect regulates crystal packing, thus enhancing the electron mobility.[41–45] Moreover, it simultaneously exhibits high ion selectivity and efficient ion transport capability under confined conditions.[46–48] In organic field-effect transistors (OFETs), by shortening the L to achieve the spatial confinement effect, higher current density and faster switching speed can be realized.[49] However, due to high operating voltages, shortening the L will trigger short-channel effects. This leads to a drastic increase in off-state current and degradation of subthreshold swing, rendering this strategy challenging to meet the reliability requirements of OFETs.[50] In contrast, OECTs are driven by low voltage (<1 V), and their working mechanism relies on the bulk doping/dedoping of the transistor channel by ions in the electrolyte, enabling OECT to maintain good charge regulation ability even in a short channel.[24,51–53] Based on this, we propose the "channel confinement effect " to elucidate how the elimination of batch-to-batch variations is achieved when the L is reduced to the nanoscale.

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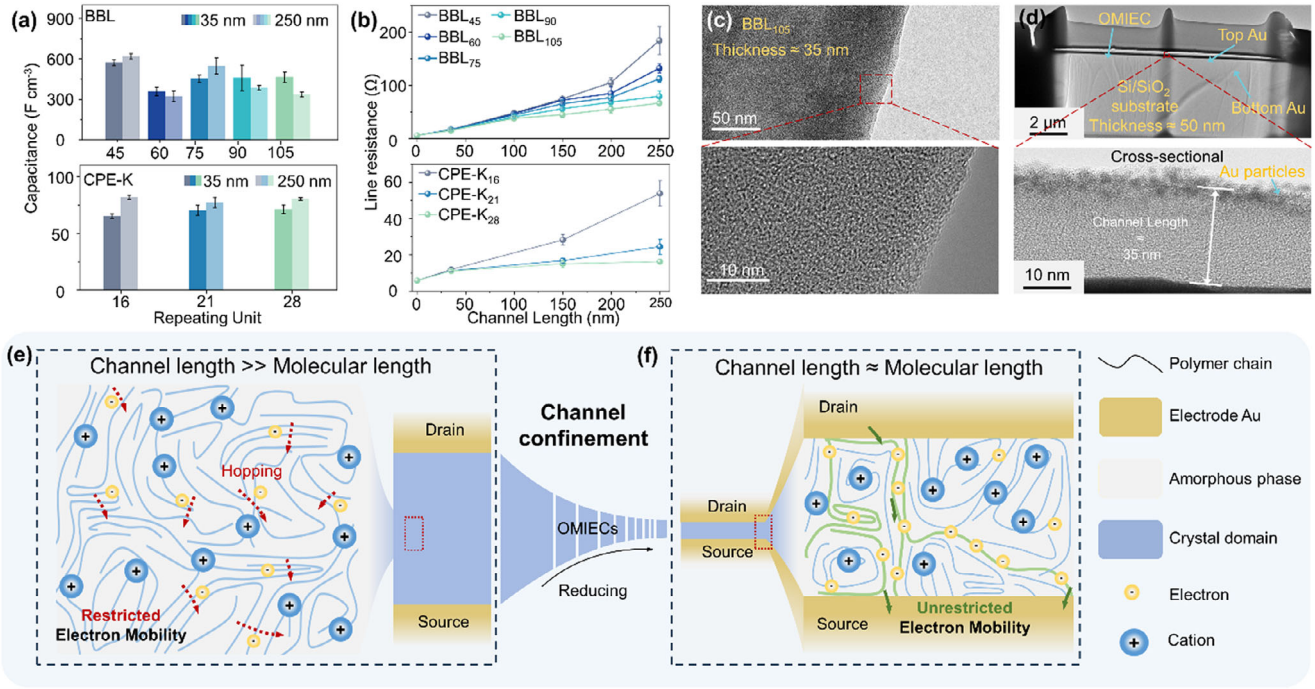


Figure 4. Physicochemical properties and film morphology characterization. a) Capacitance of BBL and CPE-K thin films based on EIS measurement. b) Contact series resistance of BBL and CPE-K-based OECTs. c) Top view and d) Cross-sectional HRTEM images of BBL_{105} thin film with a thickness of 35 nm. e,f) Schematic diagram of channel length affecting charge transport, where channel length is (e) much longer than molecular length, or f) equal to (less than) molecular length.

When L is much larger than the molecular length, μ is determined by the intercrystalline distance. As molecular chain length increases, chain entanglements in the amorphous region gradually transition to tie-chain, resulting in a macroscopic increase in μ (Figure 4i).[12,16] However, further increases in M_W can lead to a plateau or even a decline in charge mobility due to restrictions on intergranular transport. When L is approximately equal to the molecular length, chain entanglement in amorphous regions can be replaced by intramolecular transport (Figure 4j). Under these conditions, regardless of M_w, a complete molecular bridge forms within the nanoscale channel. It extends previous research by showing that the M_w and L both play key roles in efficient charge transport, with short L fostering the formation of molecular-bridges that enable rapid charge transfer via intramolecular transport. For BBL, the shortest chain length of 53.67 nm exceeds L (35 nm), whereas CPE-K's shortest chain length of 19.58 nm necessitates interchain hopping transport. Consequently, the molecular weight effect is less mitigated in CPE-K compared to BBL.

Notably, OECT performance, particularly for BBL-based devices, exhibits mild sensitivity to surface roughness. Even with a film thickness variation of 35 ± 10 nm, high-performance uniformity is maintained. The localized interchain bridge formation within confined geometries ensures that intrachain charge transport dominates, significantly reducing reliance on long-range, roughness-dependent interchain connectivity.[40,54]

Meanwhile, Pg2T-T polymers with high crystallinity are also investigated. Subsequently, OECTs with channel lengths of 50, 100, 150, 200, and 250 nm are fabricated (Figures S29–S32, Supporting Information). Its performance is almost unaffected by molecular weight and film thickness. This stands in marked contrast to the behavior observed for the BBL or CPE-K polymer described earlier. These findings indicate that for low-crystallinity polymers, restricting the channel length to the molecular chain scale can effectively enhance device performance while eliminating the molecular weight effect and batch-to-batch variations, whereas for high-crystallinity polymers, the crystalline regions reduce the impact of channel length on charge transport.

2.5. Organic Complementary Inverters

The organic complementary inverters, as one of the fundamental logic gates in organic circuits, are crucial for the development of organic circuits and exploring the application potential of organic materials in electronic devices, owing to their low fabrication cost, excellent compatibility with flexible substrates, and low power consumption.[55–58] We successfully eliminated batch-to-batch performance discrepancies in single-component polymer OECTs by leveraging the channel confinement effect. We prepare complementary inverters using BBL of different molecular weights to explore the potential of the channel limitation strategy in practical applications. Since the solvents for CPE-K and BBL are not orthogonal (Figure S52, Supporting Information), we chose another P-type material, gDPP-g2T, whose performance matches that of BBL, to prepare the organic complementary inverter, and the fabrication process is detailed in the supporting information (**Figure 5a**). A channel geometry of channel width (W)/ nominal depth (d) = 40 μm/40 μm is used for p-type and 80 μm/40 μm for n-type OECTs (Figure 5b). Patterned gDPP-g2T

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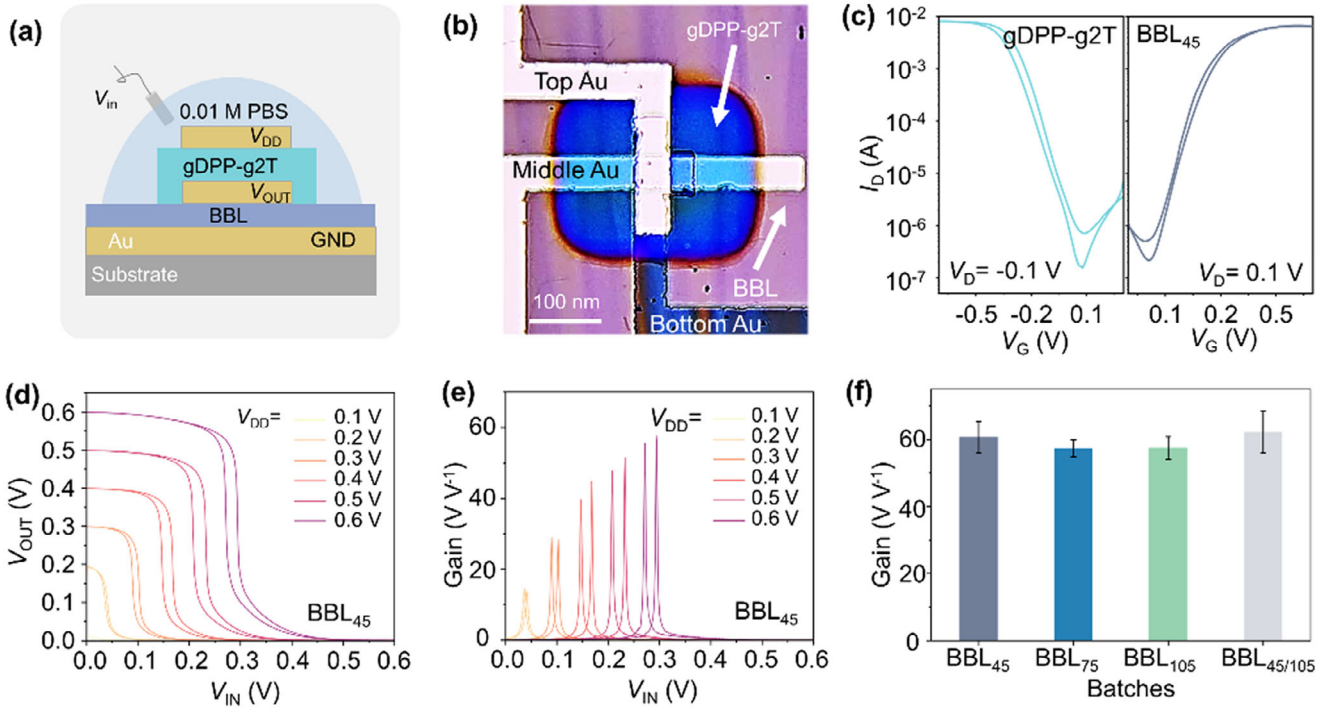


Figure 5. vOECT-based complementary inverters. a) Schematic illustration and b) optical image of the complementary inverter of BBL and gDPP-g2T. c) Transfer curves of individual vOECTs using BBL_{45} and gDPP-g2T as the channel, respectively. d) Typical voltage transfer characteristics of the inverter at supply voltages. e) Gains are calculated from the slope of the voltage transfer characteristics. f) Comparison of the voltage gain of inverters with various batches of BBL. All experimental data are collected from at least five samples.

shows a matched I_{ON}/I_{OFF} of 10^4, low V_{ON} of 0.08 V, and $g_{m,peak}$ of 40.76 mS (under a drain voltage (V_D) of −0.10 V and a gate voltage (V_G) of −0.70 V) (Figure 5c). It shows complementary inverters operating at supply voltages (V_{DD}) of 0.60 V, voltage gain over for BBL_{45}/gDPP-g2T of 60.67 V V^{-1} and BBL_{105}/gDPP-g2T of 57.46 V V^{-1} (Figure 5d,e; Figure S53, Supporting Information), respectively. Notably, at a L of 35 nm, all batches of inverter showed no significant performance gap (Figure 5f; Figure S54, Supporting Information). These results indicate that OECTs with ultra-short channels can help to eliminate the difference between batches and promote the large-scale preparation and industrialization of polymer-based OMIECs.

Building upon this foundation, we utilized organic complementary inverters to measure and amplify three types of human bioelectrical signals: electrooculography (EOG), electromyography (EMG), and electrocardiography (ECG) (**Figure 6a**). For instance, the EOG signal was amplified by a factor of 49.31 using the BBL_{45}/gDPP-g2T combination, while a similar gain of ≈44.93 was achieved with the BBL_{105}/gDPP-g2T device, demonstrating excellent inter-batch consistency (Figure 6b,c; Figure S55, Supporting Information). Inverters from all fabrication batches consistently enabled high-fidelity measurement and amplification of these signals, delivering stable and reproducible voltage gains (Figure 6d; Figures S56 and S57, Supporting Information). These results confirm that the introduction of the channel confinement effect effectively reduces batch-to-batch variations in polymer semiconductors, thereby enabling highly reproducible electrical performance across different PSC batches. We believe this stability is critical for the commercial application of organic electronics, particularly in health monitoring scenarios requiring multi-cycle and high-frequency testing for large populations (Figure 6e).

3. Conclusion

In summary, through device engineering, we utilized vertical architecture to shorten the L in OECTs from 10 µm to 35 nm. This reduction renders the L comparable to the scale of the shortest molecular chains, at which point the performance variations among BBL with different M_v (ranging from 45 to 105) become negligible. This trend is also observed in another OMIEC material, CPE-K. As the L is 35 nm, vOECTs demonstrated exceptional performance with $g_{m,peak}$ of 74 mS and I_{ON}/I_{OFF} of 10^6 for BBL, and $g_{m,peak}$ of 120 mS and I_{ON}/I_{OFF} of 10^4 for CPE-K, respectively. To investigate the factors contributing to this result, the different batches of BBL are systematically analyzed. The GIWAXS characterization and UV–vis–NIR spectra of the thin film have demonstrated that crystallinity and disorder are not the main factors affecting ionic-electronic transport. The device engineering and HRTEM test indicate that the relative length between the L and the molecule chain is a critical determinant of performance. Based on this, the channel confinement effect is proposed: reducing L is reduced to a scale comparable to or shorter than the length of the shortest molecular chain, the possibility of forming a molecular-bridge among crystalline domains, while sufficient intercrystallite connectivity makes the batch-to-batch variations no longer influence device performance. On this basis, a vertically stacked complementary inverter composed of BBL

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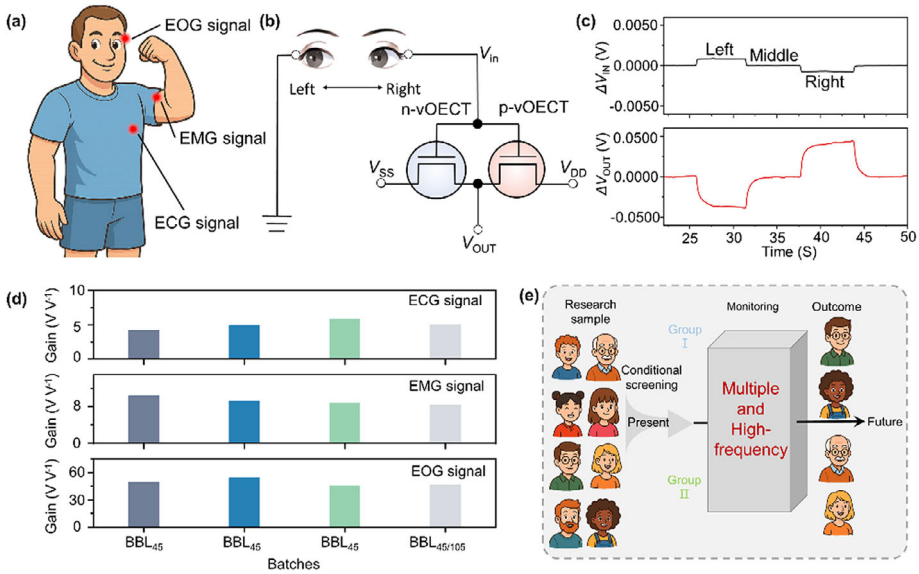


Figure 6. Applications based on complementary inverters. a) Schematic diagrams of typical human physiological electrical signals: electrooculogram (EOG), electromyogram (EMG), and electrocardiogram (ECG). b) Schematic illustration of an OECT-based complementary inverter circuit for EOG signal amplification. c) Comparison of the original and amplified EOG signals. d) Evaluation of amplification performance for three types of physiological signals using complementary inverters fabricated from different BBL batches. e) Schematic illustration of large-scale, high-frequency health monitoring across populations.

and gDPP-g2T also shows a stable voltage gain of 60.00 V V^{-1} across different molecular weights. Compared to the past approach of obtaining high-performance OECTs by adjusting the M_v of OMIECs, this work reveals the root causes of the molecular weight effect and provides guidance for developing high-performance and reproducibility devices based on low molecular weight polymer, and it lays the foundation for the future large-scale application of OECTs. In the future, this device engineering approach will be employed to address more semiconducting polymers affected by batch-to-batch variations.

4. Experimental Section

Materials: 1,2,4,5-tetraaminobenzene tetrahydrochloride (TABH) is obtained from Bidepharm. 2,6-dibromo-4,4-bis-potassium butanylsulfonate-4H-cyclopenta-[2,1-b;3,4-b']-dithiophene (provide 1) is obtained from Accela. 4,7-Bis(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)-2,1,3-benzothiadiazole (provide 2), 2,5-bis(trimethylstannyl)thiopheneis and 5,5′-Dibromo-3,3′-bis(2-(2-(2-methoxyethoxy)ethoxy)ethoxy)-2,2′- bithiophene obtained from SunaTech Inc. Methanesulfonic acid (MSA), poly (phosphoric acid) (PPA), isopropyl alcohol, $Pd_2(dba)_3$, P(*o*-tolyl)$_3$ and deuterium reagents are purchased from Adamas. Methanol and other reagents are obtained from Chron Chemicals. These compounds are used as received. 1,4,5,8-naphthalenetetracarboxylic dianhydride (NDA) is purified by dehydration. Supplementary information contains detailed information about molecular synthesis and characterization.

Synthesis Methods: All the chemicals and solvents are used as received from commercial sources without further purification unless otherwise indicated. BBL and CPE-K are prepared by the reported procedure. In brief, BBL is synthesized during high-temperature polycondensation by controlling the ratio of raw materials TABH and NDA in PPA. CPE-K is synthesized via Suzuki-Miyaura polymerization reactions of provide 1 and provide 2. Pg2T-T are synthesized as reported in the literature.[59,60] All the reactions are carried out under a nitrogen atmosphere, and all the reactions that require heating are carried out with Hei-Tec from Heidolph as a heating source.

Materials Characterization: NMR data are collected on Bruker AS-CENDTM. ^{1}H NMR and ^{13}C NMR: chemical shifts are recorded in ppm relative to tetramethyl silane and internally referenced to the residual solvent signal (for ^{1}H NMR: $CDCl_3$ = 7.26 ppm; for ^{13}C NMR: $CDCl_3$ = 77.16 ppm).

Gel Permeation Chromatography (GPC): Molecular weight distributions were determined using an Agilent PL-GPC 50 system with UP H_2O as the eluent.

Film Thickness Measurement: The film thicknesses are determined by the Atomic Force Microscope (Bruker's Innova) and Ellipsometer (Dektak XT).

Optical Microscopy: Optical images were obtained using a Nikon LV100ND microscope under ambient conditions.

UV–Vis–NIR Absorption Spectroscopy: Absorption spectra are collected using a UNICO UV-4802 UV–Vis–NIR spectrophotometer operating in wavelength scanning mode. Baseline correction was performed using a cuvette or quartz glass slide before sample measurement.

Spectroelectrochemistry Measurements: Differential spectroelectrochemistry is performed on different thick films of BBL_{75}, on an indium tin oxide glass substrate, and 0.01 M PBS aqueous solution was used as the electrolyte, with the voltage applied to the Ag/AgCl pellet electrode through a Function Generator (AFG 31 052, Tektronix). For each voltage step, a constant voltage was maintained for 200 s to ensure that the electrochemical doping reached a steady state.

CV: CV measurements are conducted in 0.1 M n-Bu_4NPF_6 in acetonitrile solution with a three-electrode configuration using an electrochemical analyzer (Metrohm, μstat-i400).

AFM: AFM images are acquired using a Bruker Innova system in tapping mode under ambient conditions.

GIWAXS: GIWAXS measurements were carried out using an XEUSS SAXS/WAXS system (Xenocs). Samples were prepared on Au-coated substrates, and the angle of incidence was set to 0.18°.

EIS: EIS measurements are conducted to obtain the volumetric capacitance (C*) in 0.01 M PBS aqueous solution with a three-electrode configuration using an electrochemical analyzer (Metrohm, μstat-i400).

FIB-HRTEM: FIB-HRTEM imaging is conducted in a focused ion beam (Helios G4 PFIB CXe) and a transmission electron microscope (JEM-F200).

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Molecular Geometry and Length Estimation: Molecular lengths are estimated via geometry optimization by mo52x/svp. Calculations were based on a syntactic chain starting from the cis-conformation for both BBL and CPE-K polymers to facilitate simplification.

Bioelectrical Signal Monitoring: ECG, EOG, and EMG tests are carried out in vitro sensing procedures (Interface 1010E). The sensors are placed externally on the body surface to detect the electrical signals generated by the eyes, heart, and muscles, respectively. All experiments assessing the skin compatibility of the wearable fabric devices are conducted on the first author, Zhu Chen, with the devices confirmed to be nonirritating and biocompatible. No ethical or safety concerns are associated with the testing.

Device Preparation and Characterization—pOECTs Fabrication and Characterization: The process consists of covering the mask template on the surface of the silicon dioxide substrate using the template method, and then deposition of chromium (Cr) and gold (Au) on the surface of the silicon dioxide substrate at 3 and 50 nm, respectively, by vapor deposition. After removing the mask template, a patterned Cr/Au electrode is obtained as the source and drain, and the channel size is W/L = 100/10 µm. The polymer is dissolved in MSA and spin-coated on the patterned substrate. After removing MSA with deionized water, 35 films are obtained. To create the encapsulation layer, a mixture containing cellulose cinnamate polymer (Cin-Cell) and a photosensitive crosslinker, DtFDA, with a weight ratio of 4:1, is applied to the surface of the prefabricated device by rotating coating, rotating at 5000 rpm using a 30 µL mixture.[61] The package layer is exposed to UV light at 365 nm wavelength for 3 min to achieve patterned crosslinking. Chloroform is then used to remove un-crosslinked crosslinkers, exposing the channel layer and source/drain contact areas.

Device Preparation and Characterization—vOECTs Fabrication and Characterization: The bottom electrode is manufactured using the same process as the plane structure, consisting of a 3 nm Cr layer and a 150 nm Au layer. MSA solution (5–20 mg mL^{-1}) is used for spin coating 35–250 nm BBL films. The top electrode is positioned according to the mask template, and then a layer of 150 nm thick Au is deposited by thermal evaporation as the top electrode. The device is encapsulated by spinning a mixture containing a photosensitive crosslinker with the same strategy, and the overlap area of the electrodes is set to 45 µm in width and length. To perform the test, Ag/AgCl is used as a gate electrode and inserted vertically into the 0.01 M PBS electrolyte. Stability measurements are conducted under a bias of $V_G = -0.30/0.60$ V ($V_D = 0.10$ V), with a test frequency of 2.50 Hz.

Device Preparation and Characterization—Vertical Stacked Complementary Inverters Fabrication and Characterization: A patterned bottom electrode was prepared by placing a shadow mask on a clean SiO_2 substrate, followed by sequential vacuum evaporation of 3 nm Cr and 150 nm Au. After removing the mask, the BBL-MSA solution was spin-coated onto the bottom electrode. The substrate was then rinsed with deionized water and annealed at 200 °C, resulting in a 35 nm BBL film. Next, a middle electrode was formed by evaporating 100 nm of gold through a shadow mask. Subsequently, a chloroform solution containing gDPP-g2T: DtFDA was spin-coated and exposed to 365 nm light for 10 min to induce photopatterning, producing a 120 nm thick film. Finally, the top electrode was completed by thermal evaporating 100 nm of Au through the mask.

Statistical Analysis: All experimental data were collected from at least three independently fabricated and measured samples ($n \geq 3$), and the sample size information for each specific figure is provided in the corresponding legend. All data processing and statistical analyses were performed using Origin 2022 (OriginLab, USA). The results are presented as mean ± standard deviation (SD).

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

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Conflict of Interest

The authors declare no conflict of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Keywords

batch variation, channel confinement effect, conjugated polymers, organic electrochemical transistors, vertical architecture

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[1] L. Zhang, J. Dong, F. Ding, *Chem. Rev.* **2021**, *121*, 6321.
[2] J. Chen, W. Huang, D. Zheng, Z. Xie, X. Zhuang, D. Zhao, Y. Chen, N. Su, H. Chen, R. M. Pankow, Z. Gao, J. Yu, X. Guo, Y. Cheng, J. Strzalka, X. Yu, T. J. Marks, A. Facchetti, *Nat. Mater.* **2022**, *21*, 564.
[3] M. Divya, N. Cherukupally, S. K. Gogoi, J. R. Pradhan, S. K. Mondal, M. Jain, A. Senyshyn, S. Dasgupta, *Adv. Mater. Technol.* **2023**, *8*, 2300256.
[4] H. Liu, D. Liu, J. Yang, H. Gao, Y. Wu, *Small* **2023**, *19*, 2206938.
[5] L. Gao, Q. Zhang, Y. Lai, M. Xie, C. Liu, D. Zhang, Y. Peng, L. Bai, M. Wu, L.-W. Feng, W. Huang, J. Yu, X. Yu, *Nano Energy* **2024**, *129*, 110062.
[6] Y. He, L. Huo, B. Zheng, *Nano Energy* **2024**, *123*, 109397.
[7] L.-Y. Xu, W. Wang, X. Yang, S. Wang, Y. Shao, M. Chen, R. Sun, J. Min, *Nat. Commun.* **2024**, *15*, 1248.
[8] B. D. Paulsen, K. Tybrandt, E. Stavrinidou, J. Rivnay, *Nat. Mater.* **2020**, *19*, 13.
[9] J. Tropp, D. Meli, J. Rivnay, *Matter* **2023**, *6*, 3132.
[10] Z. Sun, B. Khau, H. Dong, C. J. Takacs, S. Yuan, M. Sun, B. Mosevitzky Lis, D. Nguyen, E. Reichmanis, *Chem. Mater.* **2023**, *35*, 9299.
[11] M. Wu, K. Yao, N. Huang, H. Li, J. Zhou, R. Shi, J. Li, X. Huang, J. Li, H. Jia, Z. Gao, T. H. Wong, D. Li, S. Hou, Y. Liu, S. Zhang, E. Song, J. Yu, X. Yu, *Adv. Sci.* **2023**, *10*, 2300504.
[12] H.-Y. Wu, C.-Y. Yang, Q. Li, N. B. Kolhe, X. Strakosas, M.-A. Stoeckel, Z. Wu, W. Jin, M. Savvakis, R. Kroon, D. Tu, H. Y. Woo, M. Berggren, S. A. Jenekhe, S. Fabiano, *Adv. Mater.* **2022**, *34*, 2106235.
[13] D. K. Tran, A. Robitaille, I. J. Hai, C.-C. Lin, D. Kuzuhara, T. Koganezawa, Y.-C. Chiu, M. Leclerc, S. A. Jenekhe, *Chem. Mater.* **2022**, *34*, 9644.
[14] J. Tropp, D. Meli, R. Wu, B. Xu, S. B. Hunt, J. D. Azoulay, B. D. Paulsen, J. Rivnay, *ACS Mater. Lett.* **2023**, *5*, 1367.
[15] Y. Kuang, S. Heester, S. Shao, G. Ye, T. Yao, Z. Xie, L. J. A. Koster, J. Liu, *J. Mater. Chem. A* **2024**, *12*, 4866.
[16] S. Chae, T. Nguyen-Dang, J. Chatsirisupachai, A. Yi, R. J. Vázquez, G. Quek, V. Promarak, H. J. Kim, G. C. Bazan, T.-Q. Nguyen, *Adv. Funct. Mater.* **2024**, *34*, 2310852.
[17] S. M. Lee, K. H. Park, S. Jung, H. Park, C. Yang, *Nat. Commun.* **2018**, *9*, 1867.

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[18] H. Xiong, Q. Lin, Y. Lu, D. Zheng, Y. Li, S. Wang, W. Xie, C. Li, X. Zhang, Y. Lin, Z.-X. Wang, Q. Shi, T. J. Marks, H. Huang, *Nat. Mater.* **2024**, *23*, 695.
[19] J. C. Barnes, D. J. C. Ehrlich, A. X. Gao, F. A. Leibfarth, Y. Jiang, E. Zhou, T. F. Jamison, J. A. Johnson, *Nat. Chem.* **2015**, *7*, 810.
[20] Y. Wu, M. Ding, J. Wang, B. Zhao, Z. Wu, P. Zhao, D. Tian, Y. Ding, A. Hu, *CCS Chem.* **2020**, *2*, 64.
[21] H. Sun, M. Vagin, S. Wang, X. Crispin, R. Forchheimer, M. Berggren, S. Fabiano, *Adv. Mater.* **2018**, *30*, 1704916.
[22] A. T. Lill, D. X. Cao, M. Schrock, J. Vollbrecht, J. Huang, T. Nguyen-Dang, V. V. Brus, B. Yurash, D. Leifert, G. C. Bazan, T.-Q. Nguyen, *Adv. Mater.* **2020**, *32*, 1908120.
[23] J. Rivnay, S. Inal, A. Salleo, R. M. Owens, M. Berggren, G. G. Malliaras, *Nat. Rev. Mater.* **2018**, *3*, 17086.
[24] P. D'Angelo, S. L. Marasso, A. Verna, A. Ballesio, M. Parmeggiani, A. Sanginario, G. Tarabella, D. Demarchi, C. F. Pirri, M. Cocuzza, S. Iannotta, *Small* **2019**, *15*, 1902332.
[25] W. Huang, J. Chen, Y. Yao, D. Zheng, X. Ji, L.-W. Feng, D. Moore, N. R. Glavin, M. Xie, Y. Chen, R. M. Pankow, A. Surendran, Z. Wang, Y. Xia, L. Bai, J. Rivnay, J. Ping, X. Guo, Y. Cheng, T. J. Marks, A. Facchetti, *Nature* **2023**, *613*, 496.
[26] S. Fratini, M. Nikolka, A. Salleo, G. Schweicher, H. Sirringhaus, *Nat. Mater.* **2020**, *19*, 491.
[27] S. D. Kang, G. J. Snyder, *Nat. Mater.* **2017**, *16*, 252.
[28] K. C. Kao, in *Dielectric Phenomena in Solids* (Ed.: K. C. Kao), Academic Press, San Diego, **2004**, pp. 381–514.
[29] C.-K. Mai, H. Zhou, Y. Zhang, Z. B. Henson, T.-Q. Nguyen, A. J. Heeger, G. C. Bazan, *Angew. Chem., Int. Ed.* **2013**, *52*, 12874.
[30] H. Y. Wu, C. Y. Yang, Q. Li, N. B. Kolhe, X. Strakosas, M. A. Stoeckel, Z. Wu, W. Jin, M. Savvakis, R. Kroon, D. Tu, H. Y. Woo, M. Berggren, S. A. Jenekhe, S. Fabiano, *Adv. Mater.* **2021**, *34*, 2106235.
[31] D. A. Bernards, G. G. Malliaras, *Adv. Funct. Mater.* **2007**, *17*, 3538.
[32] O. Yaghmazadeh, F. Cicoira, D. A. Bernards, S. Y. Yang, Y. Bonnassieux, G. G. Malliaras, *J. Polym. Sci., Part B: Polym. Phys.* **2011**, *49*, 34.
[33] F. Cicoira, M. Sessolo, O. Yaghmazadeh, J. A. DeFranco, S. Y. Yang, G. G. Malliaras, *Adv. Mater.* **2010**, *22*, 1012.
[34] Y. Yan, Q. Chen, X. Wu, X. Wang, E. Li, Y. Ke, Y. Liu, H. Chen, T. Guo, *Acs Appl Mater Inter* **2020**, *12*, 49915.
[35] D. A. Koutsouras, F. Torricelli, P. W. M. Blom, *Adv. Electron. Mater.* **2023**, *9*, 2200868.
[36] T. R. Groves, in *Nanolithography* (Ed.: M. Feldman), Woodhead Publishing, Cambridge, U.K, **2014**, pp. 80–115.
[37] A. S. Gangnaik, Y. M. Georgiev, J. D. Holmes, *Chem. Mater.* **2017**, *29*, 1898.
[38] W. Zhai, J. Qi, C. Xu, B. Chen, Z. Li, Y. Wang, L. Zhai, Y. Yao, S. Li, Q. Zhang, Y. Ge, B. Chi, Y. Ren, Z. Huang, Z. Lai, L. Gu, Y. Zhu, Q. He, H. Zhang, *J. Am. Chem. Soc.* **2023**, *145*, 13444.
[39] R. J. Vázquez, G. Quek, S. R. McCuskey, L. Llanes, B. Kundukad, X. Wang, G. C. Bazan, *J. Mater. Chem. A* **2022**, *10*, 21642.
[40] K. Gu, C. R. Snyder, J. Onorato, C. K. Luscombe, A. W. Bosse, Y.-L. Loo, *ACS Macro Lett.* **2018**, *7*, 1333.
[41] Q. Wang, F. Yang, Y. Zhang, M. Chen, X. Zhang, S. Lei, R. Li, W. Hu, *J. Am. Chem. Soc.* **2018**, *140*, 5339.
[42] J. Tong, N. de Bruyn, A. Alieva, E. J. Legge, M. Boyes, X. Song, A. J. Walisinghe, A. J. Pollard, M. W. Anderson, T. Vetter, M. Melle-Franco, C. Casiraghi, *Nat. Commun.* **2024**, *15*, 2015.
[43] H. Gao, Y. Qiu, J. Feng, S. Li, H. Wang, Y. Zhao, X. Wei, X. Jiang, Y. Su, Y. Wu, L. Jiang, *Nat. Commun.* **2019**, *10*, 3912.
[44] Y. Gao, S. Liang, C. Jiang, M. Gu, Q. Zhang, A. Abdelhafiz, Z. Zhang, Y. Han, Y. Yang, X. Zhang, P. Liang, J. Li, X. Huang, *Sci. Adv.* **2025**, *11*, ads7154.
[45] J. Martín, M. Dyson, O. G. Reid, R. Li, A. Nogales, D.-M. Smilgies, C. Silva, G. Rumbles, A. Amassian, N. Stingelin, *Adv. Electron. Mater.* **2018**, *4*, 1700308.
[46] H. Qian, D. Wei, Z. Wang, *Nano Res.* **2023**, *16*, 11718.
[47] M. Wang, X. Hou, *Joule* **2023**, *7*, 251.
[48] A. R. Koltonow, J. Huang, *Science* **2016**, *351*, 1395.
[49] J. Xu, S. Wang, G.-J. N. Wang, C. Zhu, S. Luo, L. Jin, X. Gu, S. Chen, V. R. Feig, J. W. F. To, S. Rondeau-Gagné, J. Park, B. C. Schroeder, C. Lu, J. Y. Oh, Y. Wang, Y.-H. Kim, H. Yan, R. Sinclair, D. Zhou, G. Xue, B. Murmann, C. Linder, W. Cai, J. B.-H. Tok, J. W. Chung, Z. Bao, *Science* **2017**, *355*, 59.
[50] J. N. Haddock, X. Zhang, S. Zheng, Q. Zhang, S. R. Marder, B. Kippelen, *Org. Electron.* **2006**, *7*, 45.
[51] Y. Yao, W. Huang, J. Chen, X. Liu, L. Bai, W. Chen, Y. Cheng, J. Ping, T. J. Marks, A. Facchetti, *Adv. Mater.* **2023**, *35*, 2209906.
[52] H. Dai, W. Yue, *Adv. Eng. Mater.* **2024**, *26*, 2301860.
[53] H. Liao, J. Chen, L. Lan, Y. Yu, G. Zhu, J. Duan, X. Zhu, H. Dai, M. Xiao, Z. Li, W. Yue, I. McCulloch, *ACS Appl. Mater. Interfaces* **2022**, *14*, 16477.
[54] W. Zhu, X. Qiu, J. E. M. Laulainen, H.-l. Un, X. Ren, M. Xiao, G. Freychet, P. Vacek, D. Tjhe, Q. He, W. Wood, Z. Wang, Y. Zhang, Z. Qu, J. Asatryan, J. Martin, M. Heeney, C. R. McNeill, P. A. Midgley, I. E. Jacobs, H. Sirringhaus, *Adv. Mater.* **2024**, *36*, 2310480.
[55] L. Lu, X. Liu, P. Gu, Z. Hu, X. Liang, Z. Deng, Z. Sun, X. Zhang, X. Yang, J. Yang, G. Zu, J. Huang, *Nat. Commun.* **2025**, *16*, 3831.
[56] D. J. Wisniewski, L. Ma, O. J. Rauhala, C. Cea, Z. Zhao, A. Ranschaert, J. N. Gelinas, D. Khodagholy, *Nat. Commun.* **2025**, *16*, 517.
[57] Z. Chen, X. Ding, J. Wang, X. Guo, S. Shao, K. Feng, *Journal of Polymer Science* **2025**, *64*, 202423013.
[58] Y. Kuang, T. Yao, S. Deng, J. Dong, G. Ye, L. Zhang, S. Shao, Z. Zhu, J. Liu, J. Liu, *Adv. Mater.* **2025**, *37*, 2417691.
[59] Y. Dai, S. Dai, N. Li, Y. Li, M. Moser, J. Strzalka, A. Prominski, Y. Liu, Q. Zhang, S. Li, H. Hu, W. Liu, S. Chatterji, P. Cheng, B. Tian, I. McCulloch, J. Xu, S. Wang, *Adv. Mater.* **2022**, *34*, 2201178.
[60] M. Li, W. Feng, Y. Lan, Y. Sun, P. Li, J. Li, W. Yang, H. Li, J. Ding, J. Chen, *J. Mater. Chem. C* **2024**, *12*, 7935.
[61] Y. Lai, J. Cheng, M. Xie, J. Chen, G. Zhu, W. Huang, L.-W. Feng, *Angew. Chem., Int. Ed.* **2024**, *63*, 202401773.

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